

Puppet Mastery Part-11(Extensive scenarios practice Part-2)

1. PostgreSQL Server Configuration, EPP Drop-In Management & Idempotent Database Provisioning

Master:

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/database/manifests$ cat init.pp
class database{

    contain database::install
    contain database::service
    contain database::config
    contain database::database

    Class['database::install']->Class['database::config']->
    Class['database::service']->Class['database::database']

}

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/database/manifests$ cat install.pp
class database::install {
    $package = $facts['os']['family'] ? {
        'Debian' => 'postgresql',
        'RedHat' => 'postgresql-server',
        default => 'postgresql',
    }

    package {$package:
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mod...  
package {$package:  
  ensure => installed,  
}  
}  
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/databa  
se/manifests$ cat config.pp  
class database::config(  
  String $listen_address='localhost',  
  Integer $port=5432  
)  
{  
  
  $conf_dir = '/etc/postgresql/14/main/conf.d'  
  $postgresql_conf = '/etc/postgresql/14/main/postgresql.conf'  
  
  file{$conf_dir:  
    ensure=>directory,  
    owner=>'postgres',  
    group=>'postgres',  
    mode=>'0755'  
  }  
  
  file_line{'postgresql_include_dir':  
    path=>$postgresql_conf,  
    line=>"include_dir 'conf.d'",  
  }  
}
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mod...  
}  
  
file{"${conf_dir}/puppet.conf":  
  ensure=>file,  
  owner=>'postgres',  
  group=>'postgres',  
  mode=>'0644',  
  content=>epp('database/postgresql.conf.epp',  
    {  
      'listen_address' => $listen_address,  
      'port'=>$port  
    }  
  ),  
  require=>File[$conf_dir]  
}  
}  
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/databa  
se/manifests$ cat service.pp  
class database::service{  
  
  $serv = $facts['os']['family'] ? {  
    'Debian'=>'postgresql',  
    'RedHat'=>'postgresql',  
    default=>'postgresql'  
  }  
}
```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
}

service{$serv:
  ensure=>running,
  enable=>true
}
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/databa
se/manifests$ cat database.pp
class database::database(
  String $db_name='appdb',
){
  exec{"create_${db_name}":
    command=>"/usr/sbin/runuser -u postgres -- /usr/bin/createdb ${db_name}",
    unless=>"/usr/sbin/runuser -u postgres -- /usr/bin/psql -tAc \"SELECT 1 FROM
pg_database WHERE datname='${db_name}'\" | grep -q 1",
    require=>Class['database::service'],
  }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/databa
se/manifests$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/databa
se$ cd templates
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/databa

```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...
String $db_name='appdb',
){
  exec{"create_${db_name}":
    command=>"/usr/sbin/runuser -u postgres -- /usr/bin/createdb ${db_name}",
    unless=>"/usr/sbin/runuser -u postgres -- /usr/bin/psql -tAc \"SELECT 1 FROM
pg_database WHERE datname='${db_name}'\" | grep -q 1",
    require=>Class['database::service'],
  }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/databa
se/manifests$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/databa
se$ cd templates
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/databa
se/templates$ ls
postgresql.conf.epp
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/databa
se/templates$ cat postgresql.conf.epp
port = <%= $port %>

ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/databa
se/templates$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/databa
se$ w

```

Agent:

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Refreshing CA certificate  
Info: CA certificate is unmodified, using existing CA certificate  
Info: Refreshing CRL  
Info: CRL is unmodified, using existing CRL  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Info: Loading facts  
Notice: Requesting catalog from puppet-master:8140 (172.31.2.86)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1787327085'  
Notice: /Stage[main]/Database::Config/File[/etc/postgresql/14/main/conf.d/puppet.conf]/content:  
--- /etc/postgresql/14/main/conf.d/puppet.conf 2026-08-21 15:36:35.048075382 +0000  
+++ /tmp/puppet-file20260821-3907-oh84t1 2026-08-21 15:44:46.790377961 +0000  
@@ -1,3 +1,2 @@  
-listen_addresses = 'localhost'  
port = 5432
```

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo pg_lsclusters  
Ver Cluster Port Status Owner Data directory Log file  
14 main 5432 online postgres /var/lib/postgresql/14/main /var/log/postgresql/postgresql-14-main.log  
ubuntu@puppet-agent01:~$ id postgres  
uid=114(postgres) gid=122(postgres) groups=122(postgres),121(ssl-cert)  
ubuntu@puppet-agent01:~$ sudo -u postgres psql -l  
could not change directory to "/home/ubuntu": Permission denied  
List of databases  
+-----+-----+-----+-----+-----+-----+  
Name | Owner | Encoding | Collate | Ctype | Access privileges  
+-----+-----+-----+-----+-----+-----+  
appdb | postgres | UTF8 | C.UTF-8 | C.UTF-8 |  
postgres | postgres | UTF8 | C.UTF-8 | C.UTF-8 |  
template0 | postgres | UTF8 | C.UTF-8 | C.UTF-8 | =c/postgres +  
 | | | | | postgres=CTc/postgres  
template1 | postgres | UTF8 | C.UTF-8 | C.UTF-8 | =c/postgres +  
 | | | | | postgres=CTc/postgres  
(4 rows)  
ubuntu@puppet-agent01:~$ sudo -u postgres psql -d appdb -c "SELECT current_database();"   
could not change directory to "/home/ubuntu": Permission denied  
current_database  
-----
```

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ id postgres  
uid=114(postgres) gid=122(postgres) groups=122(postgres),121(ssl-cert)  
ubuntu@puppet-agent01:~$ sudo -u postgres psql -l  
could not change directory to "/home/ubuntu": Permission denied  
List of databases  
+-----+-----+-----+-----+-----+-----+  
Name | Owner | Encoding | Collate | Ctype | Access privileges  
+-----+-----+-----+-----+-----+-----+  
appdb | postgres | UTF8 | C.UTF-8 | C.UTF-8 |  
postgres | postgres | UTF8 | C.UTF-8 | C.UTF-8 |  
template0 | postgres | UTF8 | C.UTF-8 | C.UTF-8 | =c/postgres +  
 | | | | | postgres=CTc/postgres  
template1 | postgres | UTF8 | C.UTF-8 | C.UTF-8 | =c/postgres +  
 | | | | | postgres=CTc/postgres  
(4 rows)  
ubuntu@puppet-agent01:~$ sudo -u postgres psql -d appdb -c "SELECT current_database();"   
could not change directory to "/home/ubuntu": Permission denied  
current_database  
-----  
appdb  
(1 row)  
ubuntu@puppet-agent01:~$ |
```

2. Production-Grade Scheduled System Maintenance & Cron Job Automation

Master:

(next-page)

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production
GNU nano 6.2 data/nodes/puppet-agent01.yaml
access::users::user:
  vishnu:
    home: /home/vishnu
    shell: /bin/bash
    ssh_key: 'ed25519 AAAA...vish'
  hari:
    home: /home/hari
    shell: /bin/bash
    ssh_key: 'ed25519 BBBB...hari'
access::users::group_name: developers
access::sudo::group_name: developers
access::sudo::commands:
  - /usr/bin/systemctl restart nginx
apache_site::config::server_name: 'example.com'
apache_site::config::port: 80
apache_site::config::document_root: '/var/www/example'
database::database::db_password: 'vishnu'
maintenance::config::script_path: '/usr/local/bin/maintenance.sh'
maintenance::config::log_file: '/var/log/system-maintenance.log'
maintenance::cron::hour: 2

[ Read 23 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify  ^_ Go To Line
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production
maintenance/manifests$ sudo cat init.pp
class maintenance{
  contain maintenance::config
  contain maintenance::cron

  Class['maintenance::config']->Class['maintenance::cron']
}
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/modules/maintenance/manifests$ sudo cat config.pp
class maintenance::config(
  String $script_path,
  String $log_file
){

  file{$script_path:
    ensure=>file,
    owner=>'root',
    group=>'root',
    mode=>'0755',
    content=>epp('maintenance/maintenance.sh.epp',
      {
        'log_file'=>$log_file
      })
  }
}
```

```

ubuntu@puppet-master: /etc/puppetlabs/code/environments/production
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/maintenance/manifests$ sudo cat cron.pp
class maintenance::cron(
  Integer $hour,
  Integer $minute,
  String $script_path
){

  cron{'system-maintenance':
    ensure=>present,
    hour=>$hour,
    minute=>$minute,
    command=>$script_path,
    user=>'root'
  }
}
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/maintenance/manifests$ cd ..
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/maintenance$ sudo cat templates/maintenance.sh.epp
#!/bin/bash

echo "$(date) - maintenance completed out of Vishn's passion" >> <%= $log_file%>

```

Agent:

```

ubuntu@puppet-agent01: ~
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t
Info: Refreshing CA certificate
Info: CA certificate is unmodified, using existing CA certificate
Info: Refreshing CRL
Info: CRL is unmodified, using existing CRL
Info: Using environment 'production'
Info: Retrieving pluginfacts
Info: Retrieving plugin
Info: Loading facts
Notice: Requesting catalog from puppet-master:8140 (172.31.2.86)
Notice: Catalog compiled by puppet-master
Info: Caching catalog for puppet-agent01
Info: Applying configuration version '1787412077'
Notice: /Stage[main]/Maintenance::Config/File[/usr/local/bin/maintenance.sh]/ensure: defined content as '{sha256}7d825d67a0919f36bccfa61c56d7b24c32d6daa8f34ec51ba582bef01cfc9758'
Notice: /Stage[main]/Maintenance::Cron/Cron[system-maintenance]/ensure: created
Notice: Applied catalog in 0.05 seconds
ubuntu@puppet-agent01:~$ sudo crontab -l
# HEADER: This file was autogenerated at 2026-08-22 15:21:17 +0000 by puppet.
# HEADER: While it can still be managed manually, it is definitely not recommended.
# HEADER: Note particularly that the comments starting with 'Puppet Name' should not be deleted, as doing so could cause duplicate cron jobs.

```

```
ubuntu@puppet-agent01: ~
GNU nano 6.2 /tmp/crontab.MbAPBV/crontab
# HEADER: This file was autogenerated at 2026-08-22 15:21:17 +0000 by puppet.
# HEADER: While it can still be managed manually, it is definitely not recommen>
# HEADER: Note particularly that the comments starting with 'Puppet Name' should
# HEADER: not be deleted, as doing so could cause duplicate cron jobs.
# Puppet Name: system-maintenance
0 2 * * * /usr/local/bin/maintenance.sh

[ Read 6 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify  ^_ Go To Line
```

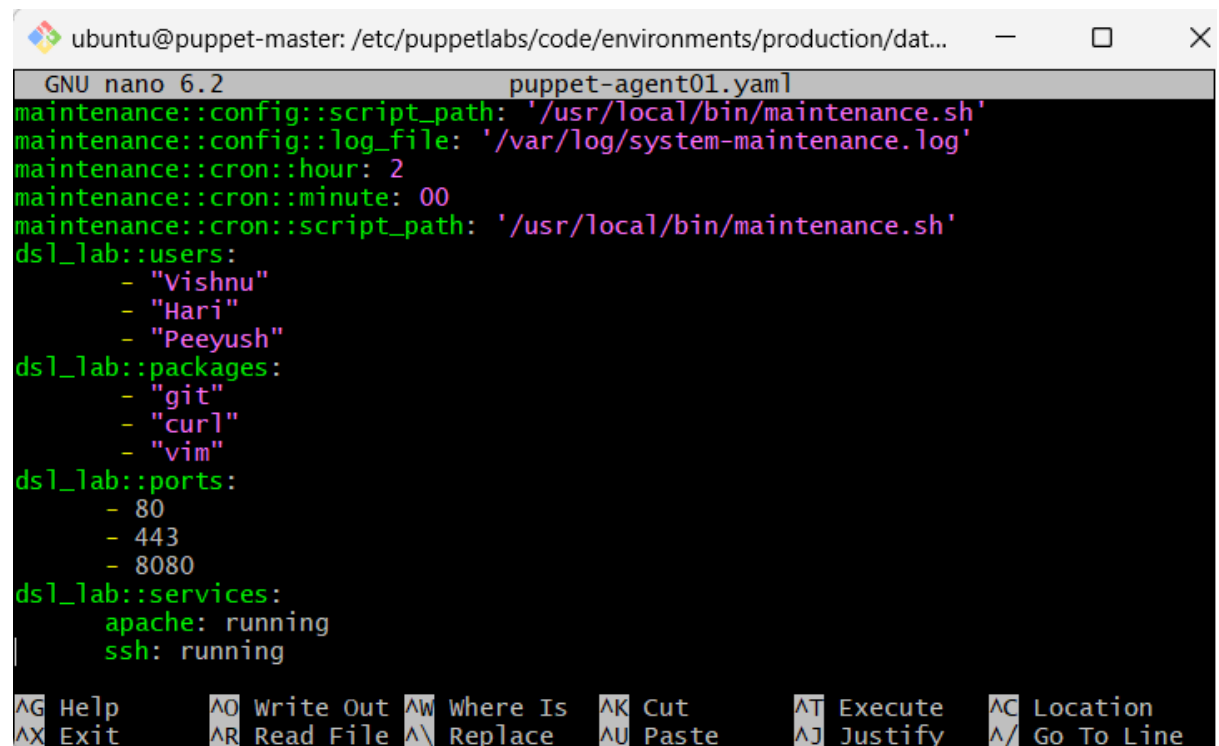
```
ubuntu@puppet-agent01: ~
ubuntu@puppet-agent01:~$ sudo crontab -e
No modification made
ubuntu@puppet-agent01:~$ sudo ls -ld /usr/local/bin/maintenance.sh
-rwxr-xr-x 1 root root 111 Aug 22 15:21 /usr/local/bin/maintenance.sh
ubuntu@puppet-agent01:~$ sudo cat /usr/local/bin/maintenance.sh
#!/bin/bash

echo "$(date) - maintenance completed out of Vishn's passion" >> /var/log/system
-maintenance.log

ubuntu@puppet-agent01:~$ sudo /usr/local/bin/maintenance.sh
ubuntu@puppet-agent01:~$ sudo cat /var/log/system-maintenance.log
Sat Aug 22 15:30:02 UTC 2026 - maintenance completed out of Vishn's passion
ubuntu@puppet-agent01:~$ |
```

3. Puppet DSL Programming, Iteration and Functional Data Manipulation

Master:



```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/dat...
GNU nano 6.2 puppet-agent01.yaml
maintenance::config::script_path: '/usr/local/bin/maintenance.sh'
maintenance::config::log_file: '/var/log/system-maintenance.log'
maintenance::cron::hour: 2
maintenance::cron::minute: 00
maintenance::cron::script_path: '/usr/local/bin/maintenance.sh'
dsl_lab::users:
  - "Vishnu"
  - "Hari"
  - "Peeyush"
dsl_lab::packages:
  - "git"
  - "curl"
  - "vim"
dsl_lab::ports:
  - 80
  - 443
  - 8080
dsl_lab::services:
  apache: running
  ssh: running
```

Help Write Out Where Is Cut Execute Location
Exit Read File Replace Paste Justify Go To Line

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/dsl_lab/manifests$ sudo cat init.pp  
class dsl_lab(  
  Array[String] $users,  
  Array[String] $packages,  
  Array[Integer] $ports,  
  Hash $services){  
  
  if $facts['os']['family'] == 'RedHat' {  
    notify{'OS check':  
      message=>'RedHat based system'  
    }  
  }  
  elsif $facts['os']['family'] == 'Debian' {  
    notify{'OS check':  
      message=>'Debian based system'  
    }  
  }  
  else{  
    notify{'OS check':  
      message=>'Other OS'  
    }  
  }  
}  
  
case $facts['os']['family']{  
  'RedHat': {
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
  
  'RedHat': {  
    notify{'OS sel':  
      message=>'RedHat selected'  
    }  
  }  
  'Debian': {  
    notify{'OS sel':  
      message=>'Debian selected'  
    }  
  }  
  default: {  
    notify{'OS sel':  
      message=>'Other OS selected'  
    }  
  }  
}  
  
unless $facts['os']['family'] == 'Windows' {  
  notify{'Platform validate':  
    message=>'Not really ur windows'  
  }  
}
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
$users.each |$user| {  
  notify{"user:${user}":  
    message=>"${user} processing"  
  }  
}  
  
$package_names= $packages.map |$package| {  
  "pkg-${package}"  
}  
  
notify{'map result':  
  message=>"${package_names}"  
}  
  
$selected_ports = $ports.filter |$port| {  
  $port>=443  
}  
  
notify{'filter result':  
  message=>"${selected_ports}"  
}  
  
$port_sum = $ports.reduce(0) |$sum,$port| {  
  $sum+$port
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
}  
  
notify{'reduce result':  
  message=>"Port sum: ${port_sum}"  
}  
  
with(80,440,8080) |$p1,$p2,$p3| {  
  notify{'with result':  
    message=>"Ports: ${p1} , ${p2} , ${p3}"  
  }  
}  
  
$range=range(1,5)  
  
notify{'range_result':  
  message=>"Range: ${range}"  
}  
  
$services.each |$name,$state| {  
  notify{"Service: ${name}":  
    message=>"Service ${name} should be ${state}"  
  }  
}  
}
```

Agent:

```
ubuntu@puppet-agent01: ~  
ubuntu@puppet-agent01:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Refreshing CA certificate  
Info: CA certificate is unmodified, using existing CA certificate  
Info: Refreshing CRL  
Info: CRL is unmodified, using existing CRL  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Info: Loading facts  
Notice: Requesting catalog from puppet-master:8140 (172.31.2.86)  
Notice: Catalog compiled by puppet-master  
Info: Caching catalog for puppet-agent01  
Info: Applying configuration version '1787486437'  
Notice: Debian based system  
Notice: /Stage[main]/Dsl_lab/Notify[OS check]/message: defined 'message' as 'Debian based system'  
Notice: Debian selected  
Notice: /Stage[main]/Dsl_lab/Notify[OS sel]/message: defined 'message' as 'Debian selected'  
Notice: Not really ur windows  
Notice: /Stage[main]/Dsl_lab/Notify[Platform validate]/message: defined 'message' as 'Not really ur windows'  
Notice: Vishnu processing  
Notice: /Stage[main]/Dsl_lab/Notify[user:Vishnu]/message: defined 'message' as '
```

```
ubuntu@puppet-agent01: ~  
Notice: Vishnu processing  
Notice: /Stage[main]/Dsl_lab/Notify[user:Vishnu]/message: defined 'message' as 'Vishnu processing'  
Notice: Hari processing  
Notice: /Stage[main]/Dsl_lab/Notify[user:Hari]/message: defined 'message' as 'Hari processing'  
Notice: Peeyush processing  
Notice: /Stage[main]/Dsl_lab/Notify[user:Peeyush]/message: defined 'message' as 'Peeyush processing'  
Notice: [pkg-git, pkg-curl, pkg-vim]  
Notice: /Stage[main]/Dsl_lab/Notify[map result]/message: defined 'message' as '[pkg-git, pkg-curl, pkg-vim]'  
Notice: [443, 8080]  
Notice: /Stage[main]/Dsl_lab/Notify[filter result]/message: defined 'message' as '[443, 8080]'  
Notice: Port sum: 8603  
Notice: /Stage[main]/Dsl_lab/Notify[reduce result]/message: defined 'message' as 'Port sum: 8603'  
Notice: Ports: 80 , 440 , 8080  
Notice: /Stage[main]/Dsl_lab/Notify[with result]/message: defined 'message' as 'Ports: 80 , 440 , 8080'  
Notice: Range: [1, 2, 3, 4, 5]  
Notice: /Stage[main]/Dsl_lab/Notify[range_result]/message: defined 'message' as 'Range: [1, 2, 3, 4, 5]'
```

```
ubuntu@puppet-agent01: ~  
pkg-git, pkg-curl, pkg-vim]'  
Notice: [443, 8080]  
Notice: /Stage[main]/Dsl_lab/Notify[filter result]/message: defined 'message' as  
'[443, 8080]'  
Notice: Port sum: 8603  
Notice: /Stage[main]/Dsl_lab/Notify[reduce result]/message: defined 'message' as  
'Port sum: 8603'  
Notice: Ports: 80 , 440 , 8080  
Notice: /Stage[main]/Dsl_lab/Notify[with result]/message: defined 'message' as '  
Ports: 80 , 440 , 8080'  
Notice: Range: [1, 2, 3, 4, 5]  
Notice: /Stage[main]/Dsl_lab/Notify[range_result]/message: defined 'message' as  
'Range: [1, 2, 3, 4, 5]'  
Notice: Service apache should be running  
Notice: /Stage[main]/Dsl_lab/Notify[Service: apache]/message: defined 'message'  
as 'Service apache should be running'  
Notice: Service ssh should be running  
Notice: /Stage[main]/Dsl_lab/Notify[Service: ssh]/message: defined 'message' as  
'Service ssh should be running'  
Notice: Service cron should be running  
Notice: /Stage[main]/Dsl_lab/Notify[Service: cron]/message: defined 'message' as  
'Service cron should be running'  
Notice: Applied catalog in 0.04 seconds  
ubuntu@puppet-agent01:~$
```

4. End-to-End Web Application Deployment, User Management and Nginx Configuration Using Hiera and EPP

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
$ ls  
manifests  templates  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
$ cd manifests  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
/manifests$ ls  
config.pp  init.pp  install.pp  service.pp  user.pp  website.pp  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
/manifests$ sudo nano init.pp  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
/manifests$ cat init.pp  
class webapp{  
  
  contain webapp::user  
  contain webapp::install  
  contain webapp::config  
  contain webapp::website  
  contain webapp::service  
  
  Class['webapp::user']->Class['webapp::install']->Class['webapp::config']->  
  Class['webapp::website']->Class['webapp::service']  
  
}
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
/manifests$ cat install.pp  
class webapp::install{  
  package{'nginx':  
    ensure=>installed  
  }  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
/manifests$ cat user.pp  
class webapp::user(  
  String $user='webapp'  
) {  
  
  user{$user:  
    ensure=>present,  
    system=>true,  
    shell=>'/usr/bin/false',  
    home=>'/var/lib/webapp',  
    managehome=>true  
  }  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
/manifests$ cat config.pp
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
/manifests$ cat config.pp  
class webapp::config(  
  String $server_name,  
  Integer $port,  
  String $document_root  
) {  
  
  file{'/etc/nginx/sites-enabled/myapp':  
    ensure=>'absent',  
    notify=>Service['nginx']  
  }  
  
  file{'/etc/nginx/sites-enabled/default':  
    ensure=>absent,  
    notify=>Service['nginx']  
  }  
  
  file{$document_root:  
    ensure=>directory,  
    owner=>'webapp',  
    group=>'webapp',  
    mode=>'0755'  
  }  
}
```

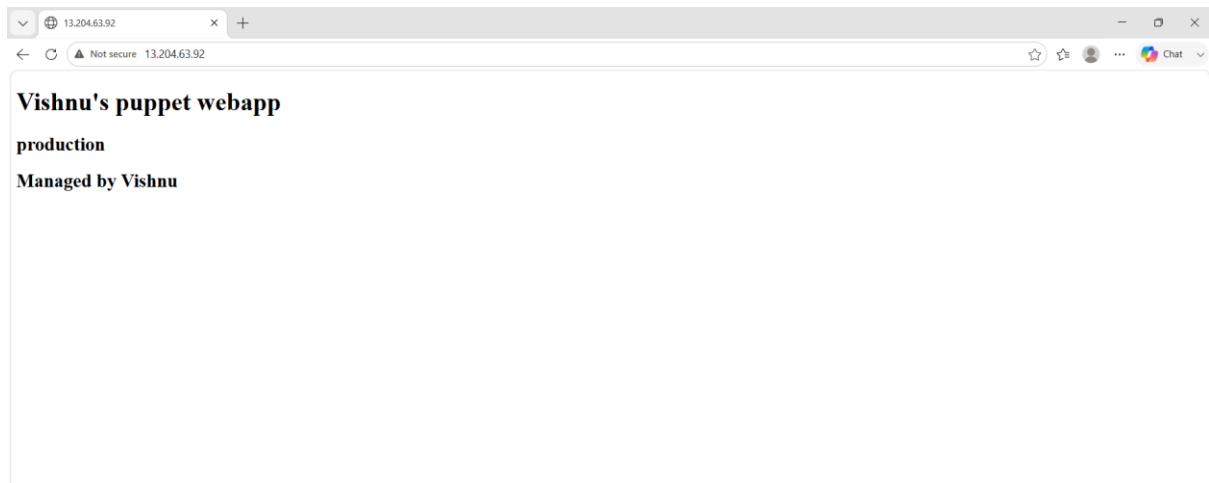
```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
file{'/etc/nginx/sites-available/webapp.conf':  
  ensure=>file,  
  owner=>'webapp',  
  group=>'webapp',  
  mode=>'0644',  
  content=>epp('webapp/vhost.conf.epp',  
    {  
      'server_name'=>$server_name,  
      'port'=>$port,  
      'document_root'=>$document_root  
    }  
  ),  
  notify=>Service['nginx']  
}  
  
file{'/etc/nginx/sites-enabled/webapp.conf':  
  ensure=>'link',  
  target=>'/etc/nginx/sites-available/webapp.conf',  
  require=>File['/etc/nginx/sites-available/webapp.conf'],  
  notify=>Service['nginx']  
}  
}
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
/manifests$ cat website.pp  
class webapp::website(  
  String $document_root,  
  String $site_name,  
  String $env  
)  
{  
  
  file{"${document_root}/index.html":  
    ensure=>file,  
    owner=>'webapp',  
    group=>'webapp',  
    mode=>'0644',  
    content=>epp('webapp/index.html.epp',  
      {  
        'site_name'=>$site_name,  
        'environment'=>$env,  
      }  
    ),  
    require=>File[$document_root]  
  }  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
/manifests$ cat service.pp  
class webapp::service{
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/mo...  
}  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
/manifests$ cd ..  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
$ cat templates/vhost.conf.epp  
server {  
  listen <%= $port %> default_server;  
  server_name _;  
  root <%= $document_root %>;  
  index index.html;  
  
  location / {  
    try_files $uri $uri/ =404;  
  }  
}  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
$ cat templates/index.html.epp  
<!DOCTYPE html>  
<h1><%= $site_name %></h1>  
<h2><%= $environment %></h2>  
<h2>Managed by Vishnu</h2>  
ubuntu@puppet-master:/etc/puppetlabs/code/environments/production/modules/webapp  
$
```

```
ubuntu@puppet-master: /etc/puppetlabs/code/environments/production/dat...  
GNU nano 6.2 puppet-agent01.yaml  
- "curl"  
- "vim"  
dsl_lab::ports:  
  - 80  
  - 443  
  - 8080  
dsl_lab::services:  
  apache: running  
  ssh: running  
  cron: running  
webapp::user::user: 'webapp'  
  
webapp::config::server_name: 'localhost'  
webapp::config::port: 80  
webapp::config::document_root: '/var/www/webapp'  
  
webapp::website::document_root: '/var/www/webapp'  
webapp::website::site_name: "Vishnu's puppet webapp"  
webapp::website::env: 'production'  
  
^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute  ^C Location  
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify  ^_ Go To Line
```

Agent:



```
ubuntu@puppet-agent01: /
ubuntu@puppet-agent01:/$ sudo /opt/puppetlabs/bin/puppet agent -t
Info: Refreshing CA certificate
Info: CA certificate is unmodified, using existing CA certificate
Info: Refreshing CRL
Info: CRL is unmodified, using existing CRL
Info: Using environment 'production'
Info: Retrieving pluginfacts
Info: Retrieving plugin
Info: Loading facts
Notice: Requesting catalog from puppet-master:8140 (172.31.2.86)
Notice: Catalog compiled by puppet-master
Info: Caching catalog for puppet-agent01
Info: Applying configuration version '1787546434'
Notice: /Stage[main]/Webapp::Config/File[/etc/nginx/sites-available/webapp.conf]
/content:
--- /etc/nginx/sites-available/webapp.conf      2026-08-24 04:33:18.437318666 +0000
+++ /tmp/puppet-file20260824-1969-9miof3        2026-08-24 04:40:35.834699958 +0000
@@ -1,7 +1,7 @@
server {
    listen 80 default_server;
    server_name _;
    root document_root;
```

```
ubuntu@puppet-agent01: /
server {
    listen 80 default_server;
    server_name _;
    root document_root;
+ root /var/www/webapp;
    index index.html;

    location / {

Notice: /Stage[main]/Webapp::Config/File[/etc/nginx/sites-available/webapp.conf]
/content: content changed '{sha256}68ff781389faa3c9a608111c21fe3d159356589d9af59
e68c9748b45e6251fc0' to '{sha256}ae906803b1582f26cf4fdc5fad81367177c7729c1e6cee
9f1baf847e79fa74f'
Info: /Stage[main]/Webapp::Config/File[/etc/nginx/sites-available/webapp.conf]:
Scheduling refresh of Service[nginx]
Notice: /Stage[main]/Webapp::Service/Service[nginx]: Triggered 'refresh' from 1
event
Notice: Applied catalog in 0.57 seconds
ubuntu@puppet-agent01:/$ curl http://localhost
<!DOCTYPE html>
<h1>Vishnu's puppet webapp</h1>
<h2>production</h2>
<h2>Managed by Vishnu</h2>
ubuntu@puppet-agent01:/$ |
```

```
ubuntu@puppet-agent01: /
<!DOCTYPE html>
<h1>Vishnu's puppet webapp</h1>
<h2>production</h2>
<h2>Managed by Vishnu</h2>
ubuntu@puppet-agent01:/$ sudo ls -l /var/www/webapp
total 4
-rw-r--r-- 1 webapp webapp 95 Aug 24 04:32 index.html
ubuntu@puppet-agent01:/$ sudo cat /etc/nginx/sites-enabled/webapp.conf
server {
    listen 80 default_server;
    server_name _;
    root /var/www/webapp;
    index index.html;

    location / {
        try_files $uri $uri/ =404;
    }
}
ubuntu@puppet-agent01:/$ sudo ls /etc/nginx/sites-enabled/webapp.conf
/etc/nginx/sites-enabled/webapp.conf
ubuntu@puppet-agent01:/$ sudo ls -l /etc/nginx/sites-enabled/webapp.conf
lrwxrwxrwx 1 root root 38 Aug 24 04:32 /etc/nginx/sites-enabled/webapp.conf -> /
etc/nginx/sites-available/webapp.conf
ubuntu@puppet-agent01:/$
```